

HORIBA
Automotive Test Systems



FTX-ONE

FTX-ONE CS

FTX-ONE Series

FTIR EXHAUST GAS ANALYZER



EMISSIONS



ELECTRIFICATION



GAV



DATA

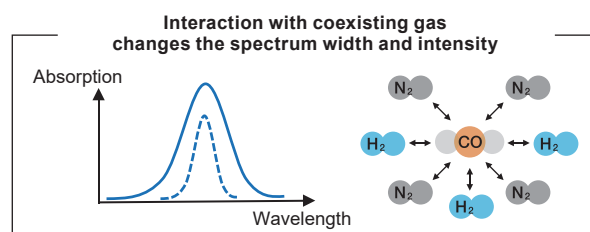
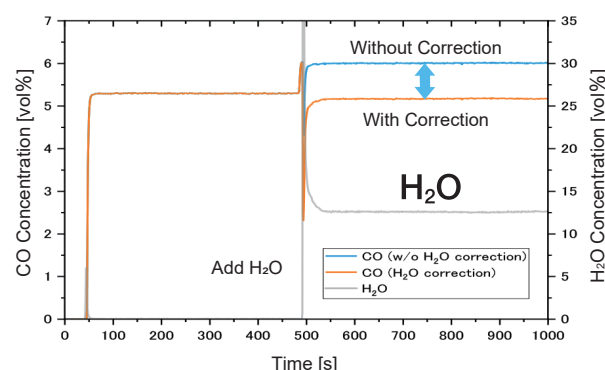
HORIBA
Automotive

Fast Response and Compact Size FTIR Analyzer for the Simultaneous Measurement of Multiple Components of Engine Exhaust Gas

HORIBA technology that supports high-precision measurement

H₂O and CO₂, which are abundant in exhaust gas, have an interference effect due to the overlap of infrared absorption spectra and a coexistence effect due to intermolecular interaction, which causes measurement error. HORIBA has clarified the correlation between interference and coexistence effects in infrared analyzers, and has developed a technology that can measure even exhaust gas containing a large amount of H₂O and CO₂ with high accuracy.

(Patent pending)



FTX-ONE Series

Direct Measurement

FTX-ONE-CS



The FTX-ONE-CS offers a 5Hz sampling rate and simultaneously measures the concentrations of multiple components including NH₃, CH₄, CO and CO₂ from direct exhaust gas.

Fast Response

The world's smallest high performance optical cell*¹ realizes a fast response time of about 1.5 seconds*² even for NH₃, which is difficult to measure.

Low Sampling Flow Rate

Achieves low sampling flow rate sampling with the world's smallest high-performance optical cell*¹ and enables measurement without affecting engine combustion or catalyst performance.

Compact Design

Approx. 1/2 the installation area and weight.*³
Improved freedom of layout and maintainability in the lab.

FTX-ONE-RS



FTX-ONE-RS is the rack type measurement system combined detector of FTX-ONE-CS and THC / O₂ detector.

Fast Response

The world's smallest high performance optical cell*¹ realizes a fast response time of about 1.5 seconds*² even for NH₃, which is difficult to measure.

Low Sampling Flow Rate

Achieves low sampling flow rate sampling with the world's smallest high-performance optical cell*¹ and enables measurement without affecting engine combustion or catalyst performance.

Fuel Consumption and A/F

Simultaneous measurement of THC and O₂ enables correlation evaluation of fuel consumption, A/F, and emission trends of each component with a single unit.

FTX-ONE-CS/RS Applications

Latest Regulations

Engine and Vehicle Development for Euro 7*⁴

For R & D applications in anticipation of the EU's next LDV regulation (measurement of NH₃, N₂O, etc.)

Catalyst

Evaluation of Catalyst and Fuel

Evaluation of aftertreatment systems such as urea SCR, gasoline three-way catalyst, and bioethanol fuel.

Latest Regulations

Measurement for EU Latest Regulations

Complaint to NH₃ concentration regulation certification under EU HDV regulation (Euro VI) and NRMM regulation (Stage V)

* 1: According to our research in May 2018

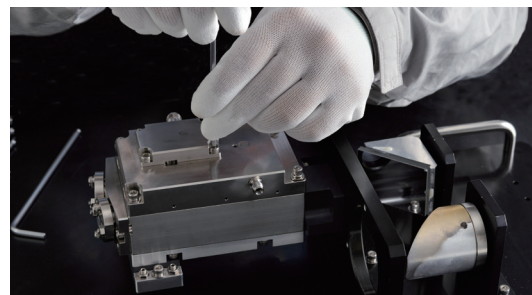
The FTX-ONE series simultaneously measures the gas concentration of multiple components by reflecting multiple lasers in the cell. By installing the world's smallest class*¹ high-performance optical cell developed in-house, high-speed response is realized with a small sample flow rate.

High Reliability

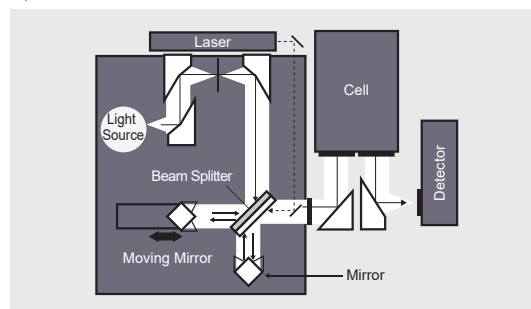
HORIBA develops and designs everything from gas sampling systems to optical interferometers in-house in order to provide highly reliable gas measuring equipment. We pay close attention to all processes and perform the best quality control to ensure long-term stable measurement.

Fourier Transform Infrared Spectroscopy (FTIR)

Fourier transform infrared spectroscopy is a method to obtain an infrared absorption spectrum by combining an interference signal generated when light divided into two optical paths is superposed again with a fast Fourier transform. Absorption spectra over a wide wavelength range can be measured every 0.2 seconds, and absorption peaks at different wavelengths can be used to simultaneously quantify multiple components in a mixed gas.



Optical Cell



Dilute Measurement

FTX-ONE-CL



The world's smallest class*¹ device that accurately measures up to 28 components such as nitrogen compounds, aldehydes, and alcohol in engine diluted exhaust gas.

Dilution measurement and mass calculation by high-precision measurement

Mass calculation is possible by evaluation in combination with a constant volume dilution sampling device and a test automation system.

Compact Design

Approx. 1/2 the installation area and weight.*³
Improved freedom of layout and maintainability in the lab.

Low Sampling Flow Rate

Minimize bag line sampling flow and easily integrate into existing systems.

FTX-ONE-CL Applications

Latest Regulations

Measurements compliant with the latest US EPA regulations*⁴

LDV Tier 3 regulation N₂O certification / NMOG certification (NMOG is NG vehicle only), Stationary engine VOC regulation NMNEHC*⁵ certification (NG engine only) is possible.

Latest Regulations

GTR No.15 compliant measurement*⁴

GTR No.15 (Global Unified Technical Regulations for LDV Exhaust Gas Testing) Compatible with measurement methods such as HCHO, C₂H₅OH, N₂O, etc.

Latest Regulations

Engine / automobile development for Euro 7*⁴

For R & D applications in anticipation of EU LDV next regulations (measurement of HCHO, C₂H₅OH, N₂O, etc.)

* 2: Response speed T₁₀₋₉₀ is about 1.5 sec on the calibration line. * 3: At the size of the footprint * 4: Please contact our sales staff for details. * 5: NMNEHC is an abbreviation for non-methane non-ethane hydrocarbon.

Outline		FTX-ONE-CS	FTX-ONE-RS	FTX-ONE-CL
	Measurement target	Exhaust gas from internal combustion engine (Direct measurement)		Exhaust gas from internal combustion engine (Dilute measurement)
	Target components*1	Max. 28 components	Max. 28 components including THC, O ₂	Max. 28 components
	Target components sets*2	Gasoline / Diesel set CNG set Ethanol set	Gasoline / Diesel set CNG set Ethanol set	Standard set : Gasoline / Diesel vehicle CNG set
	Principle	Fourier Transform Infrared (FTIR)	Fourier Transform Infrared (FTIR) Heated Flame Ionization Detection Paramagnetic Detection	Fourier Transform Infrared (FTIR)
	Sampling line temperature	113°C ± 6 °C or 191 °C ± 6 °C	113 °C ± 6 °C or 191 °C ± 6 °C	113 °C ± 3 °C
	Sampling gas flow rate	Approx. 3.5 L/min	Total Approx. 8 L/min	Continuous line: Approx. 3.5 L/min Bag line: Approx. 1.0 L/min
	Operating environment	Ambient temperature 5 - 40 °C	Ambient temperature 5 - 35 °C	Ambient temperature 5 - 40 °C
		Ambient humidity: less than 80 % relative humidity (non condensing)		
	Power requirements*3	Approx. 2.8 kVA as MAX., Approx. 2.2 kVA at stable state	Approx. 4.5 kVA as MAX., Approx. 3.5 kVA at stable state	Approx. 2.8 kVA as MAX., Approx. 2.2 kVA at stable state
	Dimensions*4	438 (W) × 661 (D) × 950 (H) mm	655 (W) × 850 (D) × 1970 (H) mm	438 (W) × 765 (D) × 950 (H) mm
	Mass*4	Approx. 150 kg	Approx. 430 kg	Approx. 160 kg
Performance	Repeatability*5 *6	Within ± 1.0 % of full scale		
	Span drift*5 *6	Within ± 1.0 % of full scale / 8 hours		
Option	Automatic replenishing device for liquid nitrogen	Max. storage amount: Approx. 12 L		
		Duration: 120 hours or more (depending on the vacuum degree of dewar)		

Components	FTX-ONE-CS/RS			FTX-ONE-CL	
	Gasoline / Diesel	CNG	Ethanol	Standard	CNG
CO	0-200 ppm			0-100 ppm	
	0-1000 ppm			0-500 ppm	
	0-5000 ppm			0-2500 ppm	
	0-2 vol%			0-1 vol%	
CO ₂	0-10 vol%			0-5 vol%	
	0-1 vol%			0-5000 ppm	
	0-5 vol%			0-2 vol%	
	0-20 vol%			0-10 vol%	
NO	0-200 ppm			0-100 ppm	
	0-1000 ppm			0-500 ppm	
	0-5000 ppm			0-2500 ppm	
NO ₂	0-200 ppm			0-100 ppm	
N ₂ O	0-200 ppm			0-20 ppm	
	—			0-100 ppm	
H ₂ O	0-24 vol%			0-15 vol%	
NH ₃	0-100 ppm			0-100 ppm	
	0-1000 ppm			0-500 ppm	
SO ₂	0-200 ppm			0-100 ppm	
HCHO	0-500 ppm*7	0-200 ppm*7	0-500 ppm*7	0-50 ppm	
HCOOH	0-100 ppm			0-50 ppm	
CH ₃ OH	—			—	0-100 ppm
C ₂ H ₅ OH	—	—	0-200 ppm	0-50 ppm	
	—	—	0-1000 ppm	0-100 ppm	
CH ₃ CHO	—	—	0-500 ppm	0-50 ppm	

Components	FTX-ONE-CS/RS			FTX-ONE-CL	
	Gasoline / Diesel	CNG	Ethanol	Standard	CNG
CH ₄	0-500 ppm			0-250 ppm	
	—	0-2000 ppm	—	—	0-1000 ppm
	—	0-1 vol%	—	—	0-5000 ppm
C ₂ H ₂	—			—	0-100 ppm
C ₂ H ₄	0-500 ppm	0-200 ppm	0-500 ppm	0-100 ppm	
	—	—	—	—	0-250 ppm
C ₂ H ₆	0-200 ppm	—	0-200 ppm	0-100 ppm	
	—	0-500 ppm	—	—	0-250 ppm
C ₃ H ₆	0-200 ppm	—	0-200 ppm	0-100 ppm	
C ₃ H ₈	—	0-200 ppm	—	—	0-100 ppm
1,3-C ₄ H ₆	0-200 ppm	—	0-200 ppm	0-100 ppm	
iso-C ₄ H ₈	0-200 ppm	—	0-200 ppm	0-100 ppm	
n-C ₄ H ₁₀	—	0-100 ppm	—	—	0-50 ppm
iso-C ₄ H ₁₀	—	0-100 ppm	—	—	0-50 ppm
C ₅ H ₆	0-500 ppm	—	0-500 ppm	0-250 ppm	
C ₇ H ₈	0-500 ppm	—	0-500 ppm	0-250 ppm	
NOx / HC sum	Sum			Sum	
THC (RS only)	0-10 ppmC - 60000 ppmC			—	
O ₂ (RS only)	0-1 vol% - 25 vol%			—	

* 1: Select when ordering from the table below

* 2: Standard combination of components and ranges, specified at time of order

* 3: With heated filter, heated hose (CS/CL : 15 m, RS : 12 m) and THC/O₂ analyzer (only RS)

* 4: Excluding protrusions * 5: Values confirmed at CO 200 ppm, CH₄ 500 ppm, and C₃H₈ 500 ppm

* 6: The performance of the span point is when a span gas with a concentration of 80% or more of the full scale range is used

* 7: Reference component

HORIBA Automotive provides innovative applications and insights, advanced automotive validation and verification technologies and equipment, and ongoing guidance and consultation to client partners across the globe. With 75 years of unmatched comprehensive scientific measurement and data management proficiencies, HORIBA Automotive's ever-expanding engineering expertise and ongoing innovations provide customers with complete solutions that solve industry challenges.

HORIBA Automotive delivers all of this from a prevailing position of lasting leadership, established integrity, recognized reliability, and continual collaboration.

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THE HORIBA GLOBAL NETWORK

ASIA

HORIBA, Ltd.
2 Miyano Higashi
Kissshoin Minami-ku
Kyoto, Japan

EUROPE

HORIBA Europe GmbH
Hans-Mess-Straße 6
61440 Oberursel
Germany
info.he@horiba.com

THE AMERICAS

HORIBA Instruments Inc.
5900 Hines Drive
Ann Arbor, MI 48108
USA
sales-ats.us@horiba.com